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Anny Dai

Software Engineer

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Software Engineer with a Master's in Applied Computing and industry experience building scalable systems in C++, Python, and cloud-native infrastructure. Proven across the full development lifecycle: from designing modular software architectures and containerized deployment pipelines to integrating computer vision components into production systems. Strong foundation in algorithms, system design, and collaborative engineering practices. Seeking Software Engineer roles to build reliable, high-impact software.

Skills

Programming	Python, C/C++, SQL, JavaScript, Bash
Frameworks & Libs	PyTorch, OpenCV, NumPy, Pandas, scikit-learn, ROS2
Systems & Infra	Docker, Linux, Git/GitHub, GCP, Databricks, REST APIs, CI/CD
Practices	System Design, Agile, Code Review, Unit Testing, Debugging, Technical Documentation

Education

Master of Science in Applied Computing (CGPA 4.0/4), *University of Toronto* — September 2024 - December 2025
Artificial Intelligence Focus — Toronto, ON, Canada

Honors Bachelor of Science with High Distinction (CGPA 3.94/4), *University of Toronto* — September 2018 - April 2024
Specialist in Computer Science, Major in Mathematics, Minor in Statistics — Toronto, ON, Canada

Technical Experience

Software Engineer Intern | Python, C++, Docker, Databricks, OpenCV, Linux — **May 2025 – December 2025**
Magna New Mobility — *Toronto, ON, Canada*

- Architected a **scalable image processing pipeline** for high-resolution industrial inputs using a windowed tiling strategy with prediction aggregation, enabling efficient large-scale inference within compute constraints.
- Designed and deployed **containerized end-to-end software pipelines** using **Docker and Databricks**, supporting reproducible data processing, model serving, and deployment workflows across environments.
- Built a **modular, multi-stage processing system** for defect analysis — combining binary segmentation, crop extraction, and multi-class classification — structured for maintainability and independent component testing.
- Improved system output quality by **+0.19 IoU** and **+0.30 Precision** through iterative algorithm tuning, data augmentation, and stitching strategy optimization.

Software Engineer Intern | C++, Python, ROS2 Foxy, OpenCV, Linux, Git — **May 2022 - June 2023**
Magna Electronics — *Brampton, ON, Canada*

- Performed **intrinsic and extrinsic camera calibration** to improve perception accuracy in vision-based sensing systems, ensuring reliable sensor alignment for downstream autonomy tasks.
- Strengthened robustness of autonomous sensing software by debugging and extending **C++/Python modules**, improving system stability under real-world operating conditions.
- Supported production readiness by managing **version-controlled customer deployments**, while separately integrating new **ROS2 Foxy C++ components on Linux** as part of the platform migration effort, maintaining code quality through testing, reviews, and GitHub workflows.

Projects

MT-GPT: Multilingual Museum Audio Guide | Python, OpenAI API, REST, Git — **September 2024 - December 2024**
- Designed and built an **end-to-end software system** for interactive multilingual museum narration, integrating the OpenAI API with a **modular prompt-control framework** to drive dynamic, context-aware content generation.
- Engineered a **configurable prompt pipeline** treating prompt components as reusable modules — separating persona, reasoning strategy, and output format — enabling rapid iteration and consistent behavior across exhibit types.
- Validated system correctness and user experience through **quantitative metrics and structured user studies**, iterating on component design based on measured readability, coherence, and satisfaction outcomes.

Generative Image Inpainting Benchmarking Framework | Python, PyTorch, Git — **September 2024 - December 2024**
- Built a **reproducible benchmarking framework** to evaluate 3 generative inpainting models across a **20k-image training set** and **5k-image test set** with 5 configurable degradation masks, structured for extensibility and experiment reproducibility.
- Implemented a **multi-metric evaluation pipeline** (MSE, MS-SSIM, FID) with automated result aggregation, enabling systematic comparison of model trade-offs across fidelity, structural similarity, and distribution alignment.
- Conducted structured failure analysis across art styles, documenting edge cases and model breakdown conditions to produce clear deployment guidance.